COMPUTER NETWORKS (CSA0711)

20 MULTIPLE CHOICE QUESTIONS – ANSWERS & JUSTIFICATIONS

1. Which TCP mechanism reduces the congestion window by half upon detecting packet loss via duplicate ACKs?

a) Slow Start
b) Fast Retransmit
c) Fast Recovery
d) Tahoe Reset

Answer: c) Fast Recovery

Why: Fast Retransmit triggers on 3 duplicate ACKs, but Fast Recovery is the mechanism that halves the congestion window.

2. A TCP sender has CWND = 8 MSS and SSTHRESH = 16. After one RTT with no loss, CWND becomes:

a) 9 MSS
b) 16 MSS
c) 12 MSS
d) 8 MSS

Answer: b) 16 MSS

Why: Since CWND (8) < SSTHRESH (16), the sender is in slow start, where CWND doubles each RTT with no loss: 8 → 16.

3. Which field in the TCP header is used to implement flow control at the receiver side?

a) Sequence number
b) Acknowledgement number
c) Window size
d) Urgent pointer

Answer: c) Window size

Why: The receiver advertises available buffer space via the Window Size field.

4. Nagle's algorithm causes increased latency primarily because it:

a) Disables ACK piggybacking
b) Buffers small segments until ACK arrives
c) Reduces CWND on every packet
d) Increases SYN backlog

Answer: b) Buffers small segments until ACK arrives

Why: Nagle's algorithm holds small segments until a pending ACK is received before sending, adding delay.

5. In a high-latency satellite link (600 ms RTT), TCP throughput is primarily limited by:

a) MTU size
b) CWND growth rate relative to RTT
c) DNS resolution time
d) UDP competition

Answer: b) CWND growth rate relative to RTT

Why: Throughput is capped by how fast CWND can grow relative to RTT (the bandwidth-delay product effect).

6. UDP is preferred over TCP for DNS queries primarily because:

a) UDP supports fragmentation
b) DNS responses are small and connection overhead is unnecessary
c) TCP cannot use port 53
d) UDP has better error correction

Answer: b) DNS responses are small and connection overhead is unnecessary

Why: Small query/response sizes make TCP's connection setup overhead unnecessary.

7. A UDP-based streaming application experiences 15% packet loss. The best application-layer mitigation is:

a) Increase UDP buffer size
b) Switch to TCP
c) Implement Forward Error Correction (FEC)
d) Reduce frame rate to 1 fps

Answer: c) Implement Forward Error Correction (FEC)

Why: FEC lets the receiver reconstruct lost data without needing retransmission, ideal for lossy real-time streams.

8. What is the maximum UDP payload size without IP-level fragmentation on a standard Ethernet link (MTU = 1500 bytes)?

a) 1480 bytes
b) 1472 bytes
c) 1460 bytes
d) 1500 bytes

Answer: b) 1472 bytes

Why: 1500 (MTU) − 20 (IP header) − 8 (UDP header) = 1472 bytes.

9. QUIC protocol uses UDP as its transport. The key advantage over TCP-based TLS is:

a) Eliminating IP header overhead
b) 0-RTT or 1-RTT connection setup
c) Removing encryption
d) Native multicast support

Answer: b) 0-RTT or 1-RTT connection setup

Why: QUIC combines transport and TLS handshakes to reduce connection setup time.

10. HTTP/1.1 persistent connections reduce latency by:

a) Enabling UDP fallback
b) Reusing the same TCP connection for multiple requests
c) Compressing HTTP headers
d) Sending requests without waiting for DNS

Answer: b) Reusing the same TCP connection for multiple requests

Why: Avoiding repeated TCP handshakes for each request reduces overhead.

11. Head-of-line blocking in HTTP/1.1 occurs when:

a) The DNS server is slow

b) A response blocks subsequent responses on the same connection
c) TCP CWND drops to zero
d) The server rejects pipelined requests

Answer: b) A response blocks subsequent responses on the same connection

Why: Responses must be returned in order, so a slow response delays all queued ones.

12. HTTP/2 multiplexing solves HOL blocking at the HTTP layer but not at the:

a) Application layer
b) DNS layer
c) TCP layer
d) IP layer

Answer: c) TCP layer

Why: All HTTP/2 streams share one TCP connection, so a lost TCP segment still blocks everything.

13. Which HTTP status code indicates that a resource has permanently moved and browsers should update bookmarks?

a) 301
b) 302
c) 304
d) 307

Answer: a) 301

Why: 301 Moved Permanently signals a permanent redirect.

14. In HTTP caching, the Cache-Control: no-cache directive means:

a) Never cache this resource
b) Validate with the server before using a cached copy
c) Cache for 0 seconds
d) Disable all caching headers

Answer: b) Validate with the server before using a cached copy

Why: no-cache does not disable caching; it requires revalidation before use.

15. Which DNS record type maps a domain name to an IPv6 address?

a) A
b) CNAME
c) AAAA
d) MX

Answer: c) AAAA

Why: AAAA records map hostnames to IPv6 addresses (A records are for IPv4).

16. A DNS resolver switches from UDP to TCP when:

a) The query type is MX
b) The response is truncated (TC bit set)
c) TTL exceeds 3600 s
d) The authoritative server is unreachable

Answer: b) The response is truncated (TC bit set)

Why: The TC bit signals the response didn't fit in a UDP datagram, prompting a TCP retry.

17. Which attack exploits the UDP-based DNS protocol by sending forged responses with a matching transaction ID?

a) ARP poisoning
b) DNS cache poisoning
c) SYN flooding
d) BGP hijacking

Answer: b) DNS cache poisoning

Why: Attackers forge responses with a guessed/matching transaction ID to poison resolver caches.

18. Setting a very low TTL (e.g., 30 s) for a DNS record primarily increases:

a) Cache hit ratio
b) Recursive resolver query load
c) DNSSEC signature validity
d) Zone transfer frequency

Answer: b) Recursive resolver query load

Why: Low TTL forces frequent re-resolution, increasing load on the recursive resolver.

19. A user browses to a new website. The correct sequence of protocol interactions for the first byte of the page is:

a) HTTP → TCP → DNS
b) DNS → TCP handshake → HTTP GET
c) TCP → DNS → HTTP
d) DNS → HTTP → TCP

Answer: b) DNS → TCP handshake → HTTP GET

Why: Name resolution must happen before a TCP connection can be opened, before HTTP can be used.

20. Under 20% packet loss, which protocol combination gives the best performance for a real-time multiplayer game?

a) TCP for game state, HTTP for assets
b) UDP with application-layer sequencing for game state, HTTP/3 for assets
c) HTTP/2 for all traffic
d) TCP with Nagle enabled for all traffic

Answer: b) UDP with application-layer sequencing for game state, HTTP/3 for assets

Why: UDP avoids TCP's HOL blocking for game state; HTTP/3 (QUIC/UDP) handles asset loading efficiently.

QUICK-REFERENCE ANSWER SHEET

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
c	b	c	b	b	b	c	b	b	b	b	c	a	b	c	b	b	b	b	b